Timothy Mitchell

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EDUCATION

University of North Carolina at Charlotte, Charlotte, NC
Bachelor of Science in Mechanical Engineering, Concentration in Motorsports Expected Graduation May 2028

Durham Tech Community College, Durham, NC
Associate of Science in Engineering May 2025

RELATED COURSEWORK:

Statics | Solids | Mechatronics 1 | Manufacturing Systems | Machine Shop | Dynamics | Thermodynamics | Mechatronics 2 | Linear Algebra | Fluid Mechanics | Mechanics of Materials | Computational Methods

SKILLS:

SolidWorks, Onshape, **Python**, MATLAB, Java, React, C, C++, Visual Studio Code, **Excel**, Microsoft Tools, Computer Hardware, 3D modeling, 3D design, Manufacturing, **3D printing, Arduino, ESP32, KiCAD, ANSYS, Mastercam, Vericut, MoTeC, Manual Mill and Lathe**

ENGINEERING EXPERIENCE

Formula Student Vehicle Dynamics Design Team Member, 49ers Racing, Charlotte, NC January 2026 – Present
- Designed/validated rockers, 40 cups, and straight cups in SolidWorks/Ansys for safety and deflection.
- Used Excel to build an ARB calculator to determine roll stiffnesses and validate design choices.
- Used MoTeC to build worksheets tracking performance targets and drive data-based setup changes.
- Built a 400+ part SolidWorks assembly of the 2027 car to validate suspension components and hardware.
- Performed an Ansys RCA on a previous car's shock bar to identify failure modes.

Manufacturing Engineering Intern, Northeast Tool and Manufacturing, Mathews, NC August 2025 – Present
- Worked in SolidWorks to develop complex fixtures for 5-axis CNC machines.
- Used Mastercam to develop 5-axis toolpaths for DMG MORI machines to produce critical fixtures.
- Participated in Kaizen events to optimize and modernize our shipping department from the ground up.
- Developed custom software tools integrating with our ERP's API to improve efficiency.
- Designed, validated, and manufactured a transformer tower in 5 days to deliver on a critical deadline.

Lead Mentor, Codewiz FTC Team, Durham, NC August 2023 – August 2025
- Led a team of 10 students to build a robot in 4 months and compete in FTC competitions.
- Taught advanced CAD principles and Java programming to students.
- Organized grants and secured over $2,000 in funding to cover 2 years of team expenses.

PROJECTS

Junior Design Project, Robotic Design Competition, Charlotte, NC January 2026 – May 2026
- Worked in a team of 7 to design, validate, manufacture, and test a robot for the competition challenge.
- Used SolidWorks topology optimization to design an ultralight chassis, the competition's first sub-one-pound robot.
- Designed the electronics and programmed the ESP32 microcontroller.
- Used 3D printing creatively to produce topology-optimized parts and functional GT3 belts out of TPU.
- Used CAD to design and assemble the completed robot, including custom double-herringbone gears.

Technical Lead, Rival Robotics Competition Team, Raleigh, NC August 2024 – Present
- Designed a 3D model in Onshape of a competition robot based on defined criteria and constraints.
- Incorporated manufacturability and machinability into the robot design.
- Developed a website using React to showcase the team's accomplishments and information.

OTHER EXPERIENCE

Center Director / Manager August 2023 – August 2025
Code Wiz Durham, Durham, North Carolina
- Led and trained a team of 10 coaches, managing scheduling and performance for team success.
- Organized and ran school partnerships, teaching robotics in classroom settings of 15+ students.